MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

SWITCH, S/D, MULTISPEED, ANALOG, KEY CODE HUH

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic S/D multispeed switch module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.529 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be HUH.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/225-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage of 5.0 ±0.5 V dc and ±15.0 ±1.5 V dc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

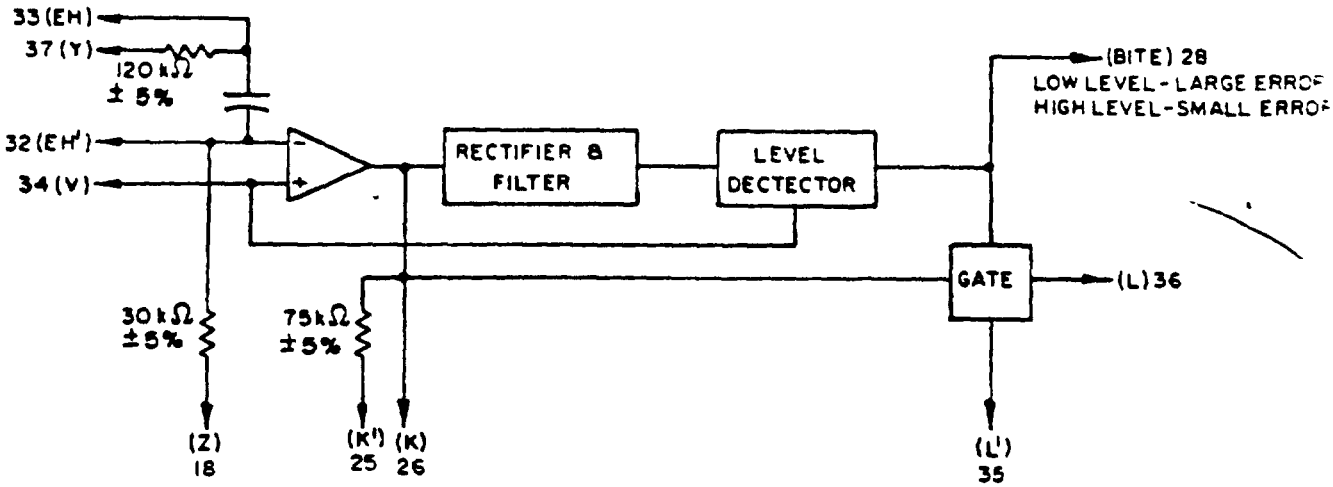
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

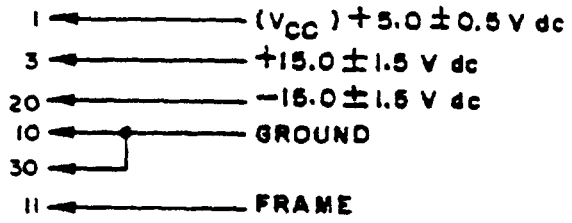
4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.




NOTE: The following are unused pins: 2, 4, 5, 6, 7, 8, 9, 12, 13, 14, 15, 16, 17, 19, 21, 22, 23, 24, 27, 29, 31, 38, 39, and 40.

*FIGURE 1. HUH function diagram, S/D multispeed switch.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.

*TABLE I. Module electrical requirements.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | Acceptance requirements 25°C ±5°C ambient temperature | | | | Operating temperature 0°C to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| A1-A26 | Isolation | II | S | 10 | - | 1 | - | 10 | - | 1 | - | MΩ |
| B1-B10 | Hot socket | III | Q | ← See table III and footnote 4 below → | | | | | | | | |
| C1,C2 | Maximum input voltage | IV | Q | ← Survival → | | | | | | | | |
| D1-D4 | Output short circuit current | V | S | ← See table V and footnote 4 below → | | | | | | | | |
| E | $V_S$ stress | VI | Q | ← Survival → | | | | | | | | |
| F1-F6 | Crossover | VII | X | ← See table VII and footnote 4 below → | | | | | | | | |
| G1 | DC bias current, crossover switch | VIII | S | 0 | 750 | 0 | 750 | 0 | 750 | 0 | 750 | μA |
| G2 | DC bias current, crossover switch | VIII | X | 2.4 | 6.6 | 2.4 | 6 5 | 2.4 | 5.5 | 2.4 | 5.5 | V |
| H1-H3 | Resistance | IX | X | See table IX and footnote 4 below | | | | | | | | |
| H4 | Continuity | IX | S | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | Ω |
| J1 | Power supply filter capacitance | X | Q | 12 | 18 | 12 | 18 | 12 | 18 | 12 | 18 | μF |
| J2,J3 | Power supply filter capacitance | X | Q | 0.08 | 0.12 | 0.08 | 0.12 | 0.08 | 0.12 | 0.08 | 0.12 | μF |
| K1-K3 | Power supply current | XI | S | 0 | 15 | 0 | 15 | 0 | 15 | 0 | 15 | mA |
| L1,L2 | BITE propagation delay time ($t_{PHL}$ and $t_{PLH}$) | XII | S | 0 | 50 | 0 | 50 | 0 | 50 | 0 | 50 | ms |
| L3,L4 | BITE transition time ($t_{TLH}$ and $t_{THL}$) | XII | S | 0.1 | 200.1 | 0.1 | 200.1 | 0.1 | 200.1 | 0.1 | 200.1 | μs |

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only
4/ Limits for initial and end-of-life for 25°C and operating temperature are the same.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.

* 4.4.2 Order of testing. Test conditions A through L shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence. If test conditions A, B, C, D, or E are performed at any time during quality assurance testing, they shall be followed by performance of all tests marked with an X in the test type column of table I.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through L in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions F through H, K, and L shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests F through H, K, and L shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests F through H, K, and L shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/225-1 supersedes part number M28787/225. The two parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

| | |
|---|---|
| Maximum power dissipation | 0.75 W (outputs not loaded) |
| Supply voltages | 5 V dc ±10 percent<br>±15 V dc ±10 percent |

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.

6.5 Module function. This module contains a switch for multispeed (1:36) S/D (synchro to digital) converters and "BITE" circuitry for functional verification used in S/D converters.

6.5.1 Input to output relationship. The error signal out of L' depends upon the error signal to EH. If the error signal to EH is $\leq 1.25^{\circ}$ then L' is the error signal from L. The "BITE" output indicates an error of $3.2^{\circ}$, or greater. The relationship for the "BITE" output is as follows:

| | |
|---|---|
| High level | Small error. |
| Low level | Large error. |

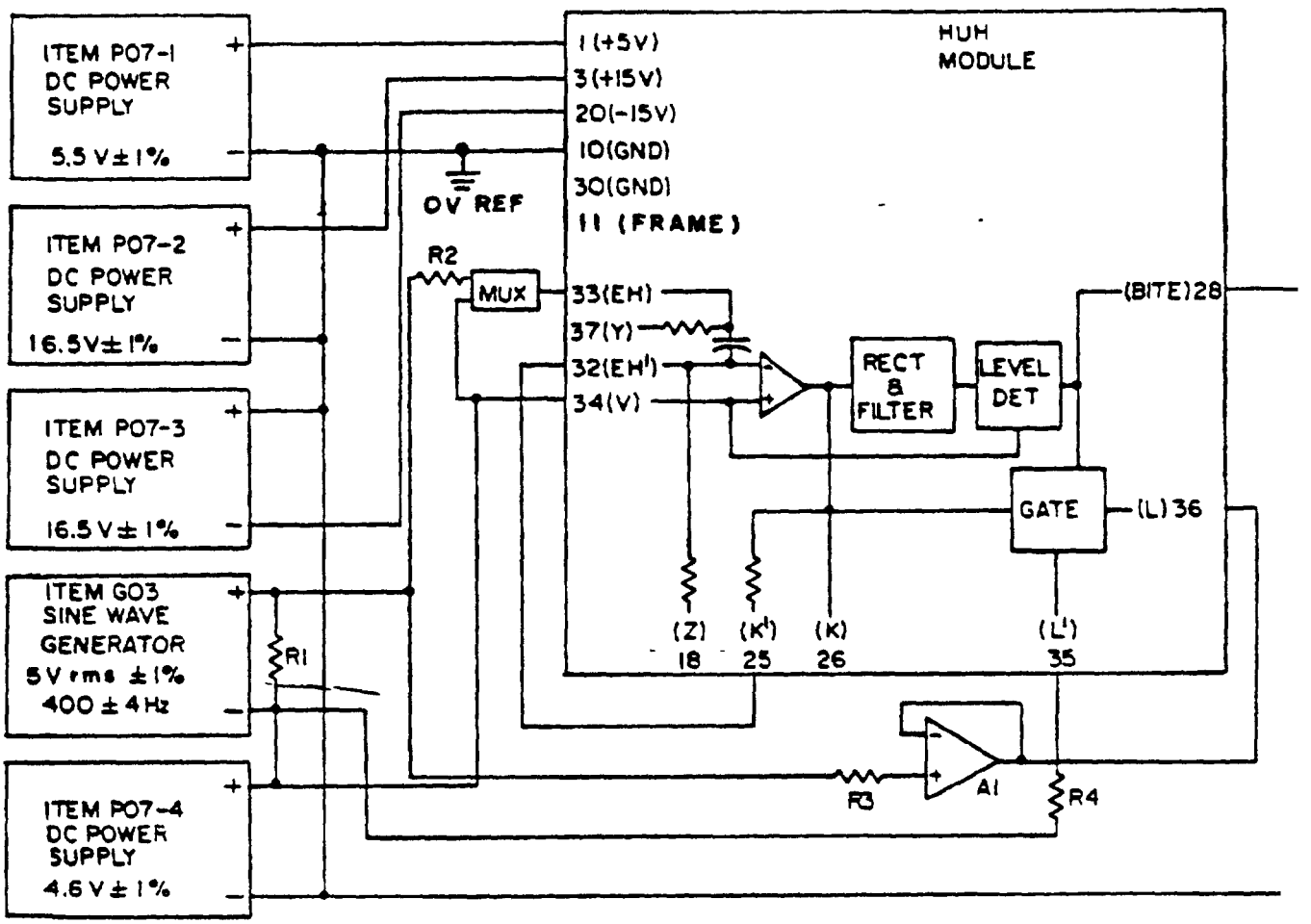
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N529)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.




NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2 = 56.2 kΩ ±1%, 1/8 W.
3. R3 = 1 kΩ ±1%, 1/8 W.
4. R4 = 20 kΩ ±1%, 1/8 W.
5. MUX = RCA CD4051A multiplexer, or equivalent.
6. Operate the MUX at 20 Hz.
7. A1 = Fairchild type μA741 OpAmp, or equivalent.

FIGURE 2. Operating temperature, component temperature, and life test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.

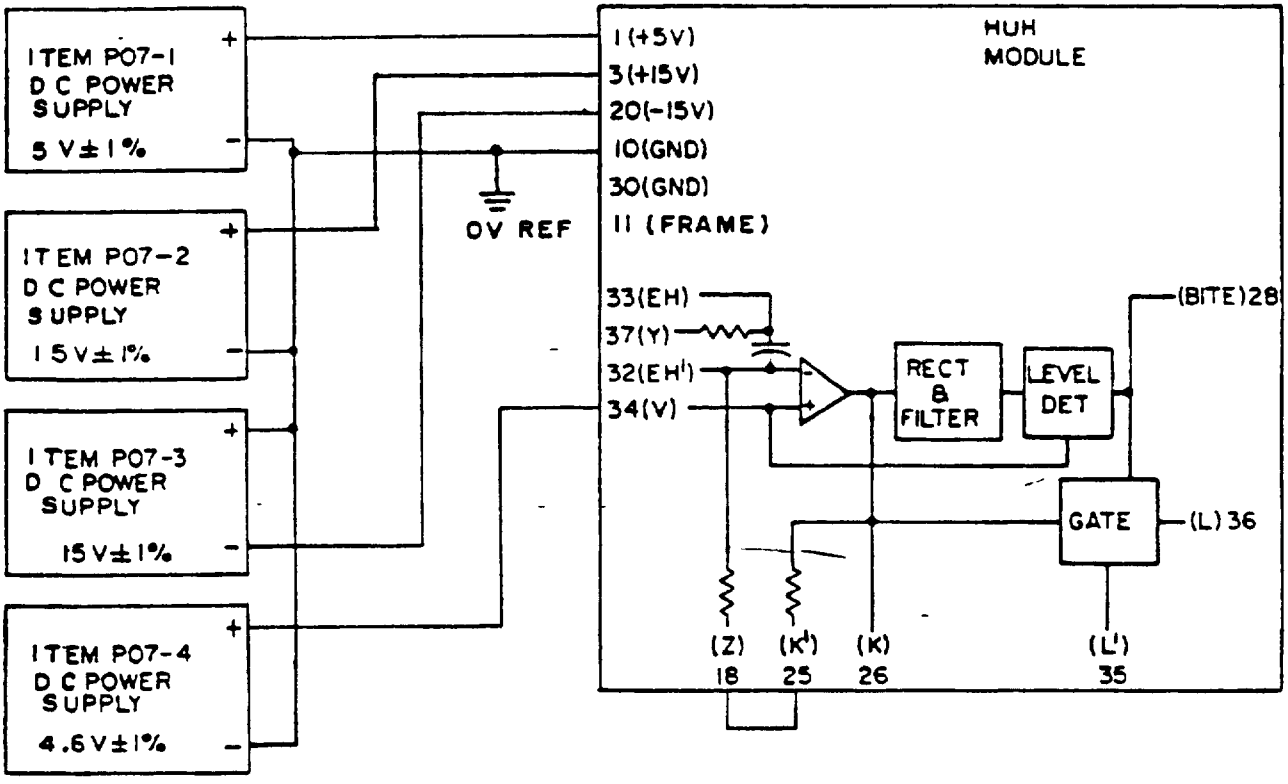



FIGURE 3. Vibration test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.

TABLE II. Isolation test.

| Test condition | Parameter | Connect | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| A1 | Isolation, pins to frame pin | 1,2,3,4,5,6,7,8,9,10, 12,13,14,15,16,17,18, 19,20,21,22,23,24,25, 26,27,28,29,30,31,32, 33,34,35,36,37,38,39, 40 | 11 | 10 | - | MΩ | 4 |
| A2 | Isolation, used to unused pins | C | 2,4,5,6,7,8,9,12,13, 14,15,16,17,19,21,22, 23,24,27,29,31,38,39, 40 | " | " | " | " |
| A3 | Isolation, unused to unused pins | 40 | 2,4,5,6,7,8,9,12,13, 14,15,16,17,19,21,22, 23,24,27,29,31,38,39 | " | " | " | " |
| A4 | " | 39 | 2,4,5,6,7,8,9,12,13, 14,15,16,17,19,21,22, 23,24,27,29,31,38 | " | " | " | " |
| A5 | " | 38 | 2,4,5,6,7,8,9,12,13, 14,15,16,17,19,21,22, 23,24,27,29,31 | " | " | " | " |
| A6 | " | 31 | 2,4,5,6,7,8,9,12,13, 14,15,16,17,19,21,22, 23,24,27,29 | " | " | " | " |
| A7 | " | 29 | 2,4,5,6,7,8,9,12,13, 14,15,16,17,19,21,22, 23,24,27 | " | " | " | " |
| A8 | " | 27 | 2,4,5,6,7,8,9,12,13, 14,15,16,17,19,21,22, 23,24 | " | " | " | " |
| A9 | " | 24 | 2,4,5,6,7,8,9,12,13, 14,15,16,17,19,21,22, 23 | " | " | " | " |
| A10 | " | 23 | 2,4,5,6,7,8,9,12,13, 14,15,16,17,19,21,22, | " | " | " | " |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.

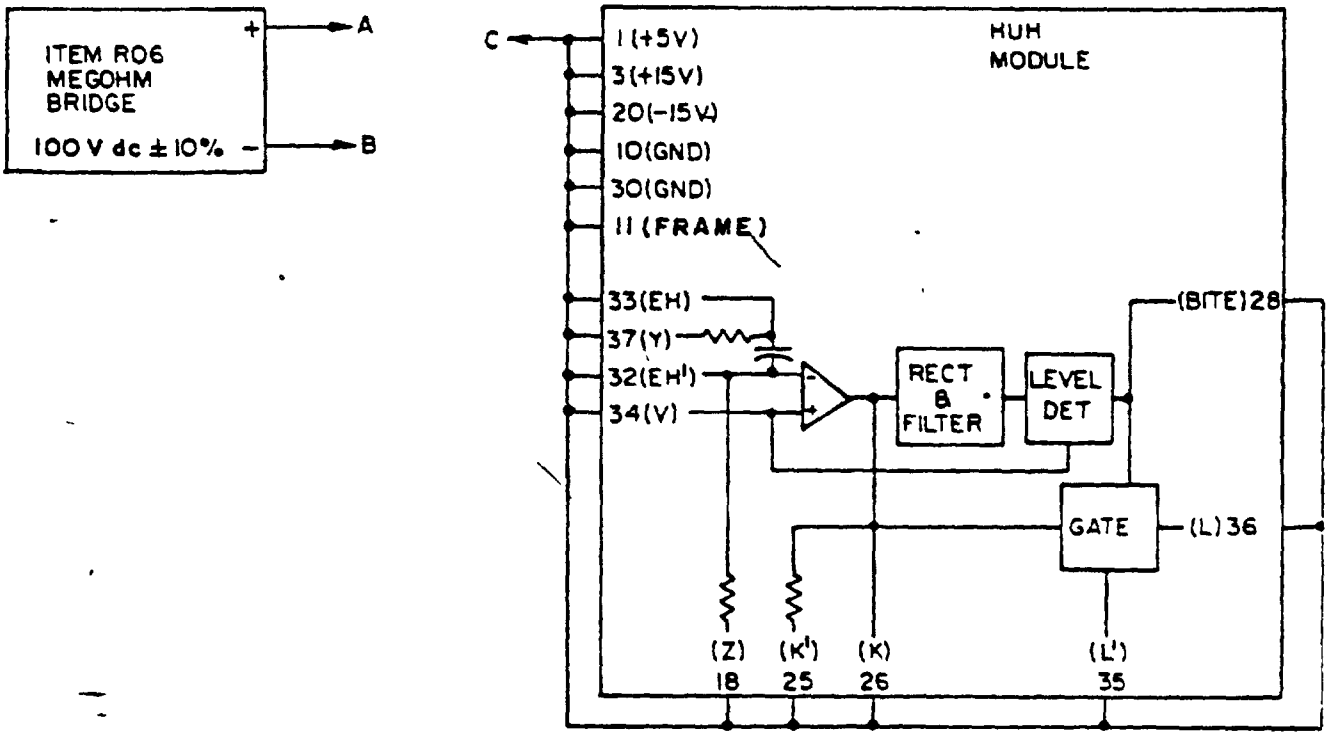
TABLE II. Isolation test - Continued.

| Test condition | Parameter | Connect | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| A11 | Isolation, unused to unused pins | 22 | 2,4,5,6,7,8,9,12,13, 14,15,16,17,19,21 | 10 | - | MΩ | 4 |
| A12 | " | 21 | 2,4,5,6,7,8,9,12,13, 14,15,16,17,19 | " | " | " | " |
| A13 | " | 19 | 2,4,5,6,7,8,9,12,13, 14,15,16,17 | " | " | " | " |
| A14 | " | 17 | 2,4,5,6,7,8,9,12,13, 14,15,16 | " | " | " | " |
| A15 | " | 16 | 2,4,5,6,7,8,9,12,13, 14,15 | " | " | " | " |
| A16 | " | 15 | 2,4,5,6,7,8,9,12,13, 14 | " | " | " | " |
| A17 | " | 14 | 2,4,5,6,7,8,9,12,13 | " | " | " | " |
| A18 | " | 13 | 2,4,5,6,7,8,9,12 | " | " | " | " |
| A19 | " | 12 | 2,4,5,6,7,8,9 | " | " | " | " |
| A20 | " | 9 | 2,4,5,6,7,8 | " | " | " | " |
| A21 | " | 8 | 2,4,5,6,7 | " | " | " | " |
| A22 | " | 7 | 2,4,5,6 | " | " | " | " |
| A23 | " | 6 | 2,4,5 | " | " | " | " |
| A24 | " | 5 | 2,4 | " | " | " | " |
| A25 | " | 4 | 2 | " | " | " | " |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.




**WARNING:**

**When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.**

**FIGURE 4.** Isolation test circuit for table II.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.

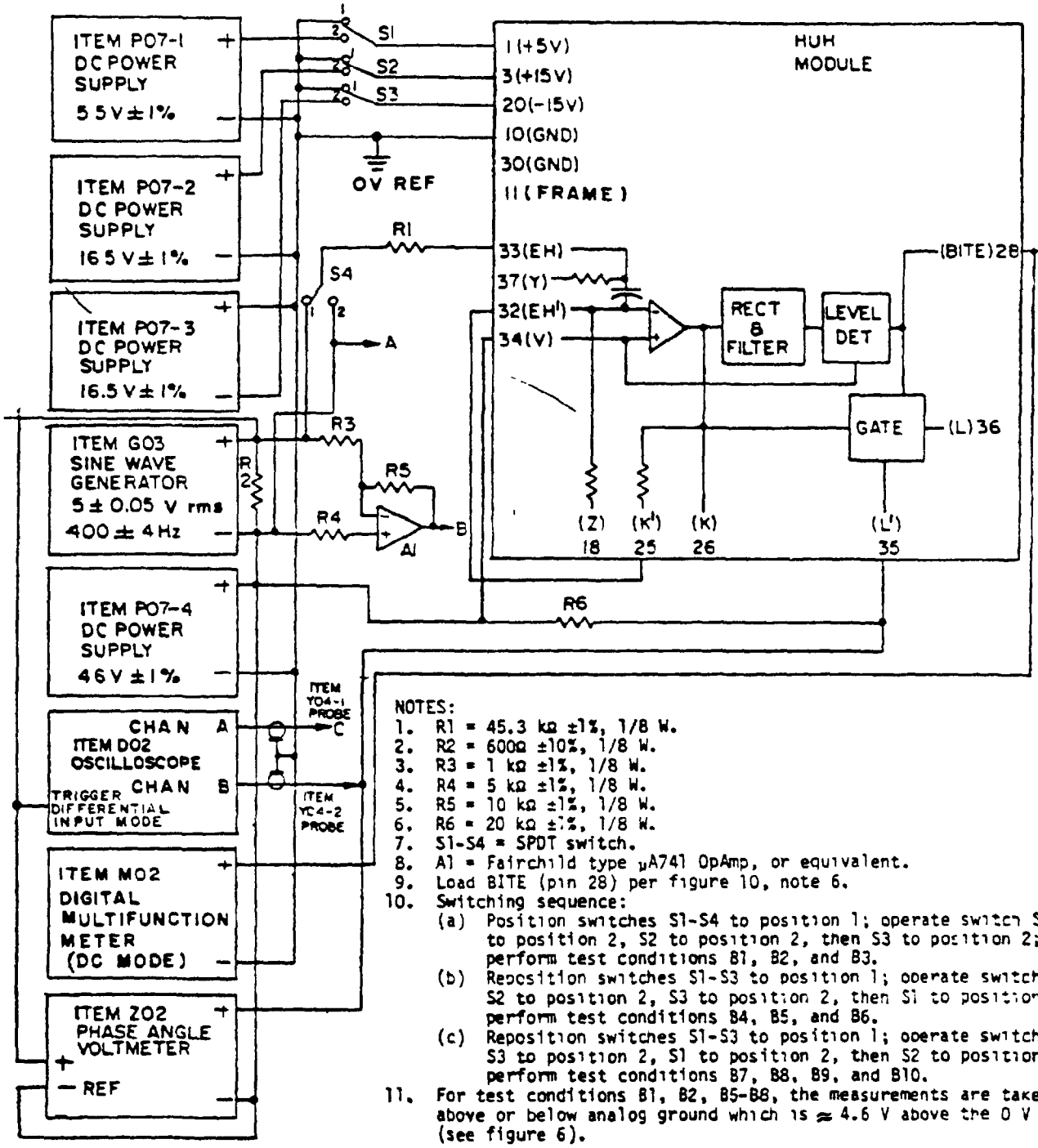
TABLE III, Hot socket test.

| Test condition | Parameter | Position switch S4 | Connect | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | Min | Max | | |
| B1 | Hot socket | 1 | 36 | Open | 34 | +4 | --- | Vpk | 5,6 |
| B2 | " | 1 | 36 | Open | 34 | -4 | --- | Vpk | " |
| B3 | " | 2 | Open | 36 | 36 | 0 | 20 | mVp-p | " |
| B4 | " | 2 | Open | 36 | 36 | 0 | 20 | Vp-p | " |
| B5 | " | 1 | 36 | Open | 34 | +4 | --- | Vpk | " |
| B6 | " | " | " | " | " | -4 | " | " | " |
| B7 | " | " | " | " | " | +4 | " | " | " |
| B8 | " | " | " | " | " | -4 | " | " | " |
| B9 | " | 2 | Open | 36 | 36 | 0 | 20 | mVp-p | " |
| B10 | " | 1 | 36 | Open | Open | 178.5 | 181.5 | Degree | " |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.

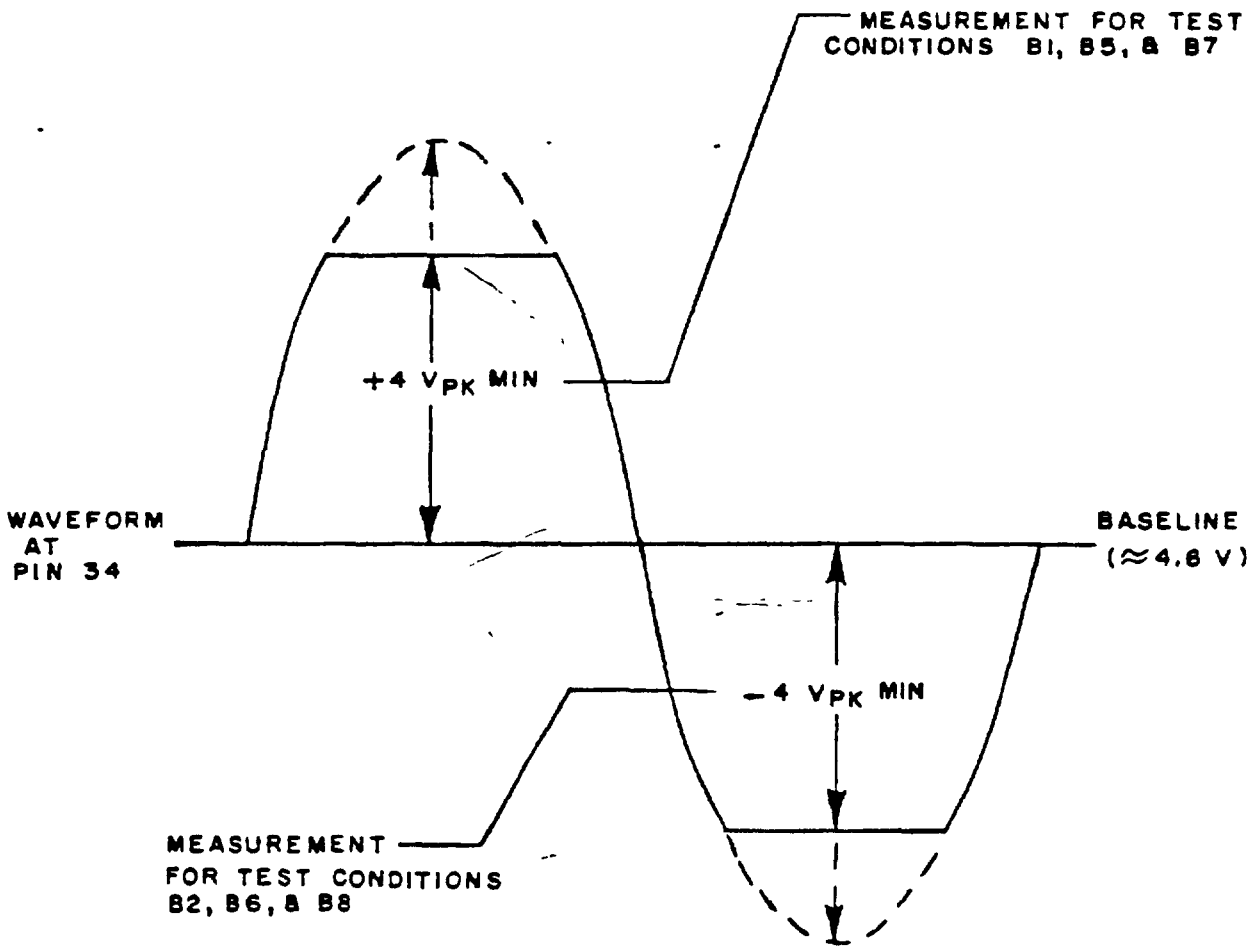



NOTES:
1. R1 = 45.3 kΩ ±1%, 1/8 W.
2. R2 = 600Ω ±10%, 1/8 W.
3. R3 = 1 kΩ ±1%, 1/8 W.
4. R4 = 5 kΩ ±1%, 1/8 W.
5. R5 = 10 kΩ ±1%, 1/8 W.
6. R6 = 20 kΩ ±1%, 1/8 W.
7. S1-S4 = SPDT switch.
8. A1 = Fairchild type μA741 OpAmp, or equivalent.
9. Load BITE (pin 28) per figure 10, note 6.
10. Switching sequence:
    (a) Position switches S1-S4 to position 1; operate switch S1 to position 2, S2 to position 2, then S3 to position 2; perform test conditions B1, B2, and B3.
    (b) Reposition switches S1-S3 to position 1; operate switch S2 to position 2, S3 to position 2, then S1 to position 2, perform test conditions B4, B5, and B6.
    (c) Reposition switches S1-S3 to position 1; operate switch S3 to position 2, S1 to position 2, then S2 to position 2, perform test conditions B7, B8, B9, and B10.
11. For test conditions B1, B2, B5-B8, the measurements are taken above or below analog ground which is ≈ 4.6 V above the 0 V REF (see figure 6).

*FIGURE 5. Hot socket test circuit for table III.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.




NOTE: The waveform at pin 34 may or may not be clipped, but must meet the requirements of table III.

*FIGURE 6. Hot socket test circuit waveform for table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.

TABLE IV. Maximum input voltage test.

| Test condition | Parameter | Position switch S1 | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | | Min | Max | | |
| C1 | Maximum input voltage | 1 | - - | - - | - - | 7 |
| C2 | Maximum input voltage | 2 | - - | - - | - - | 7 |

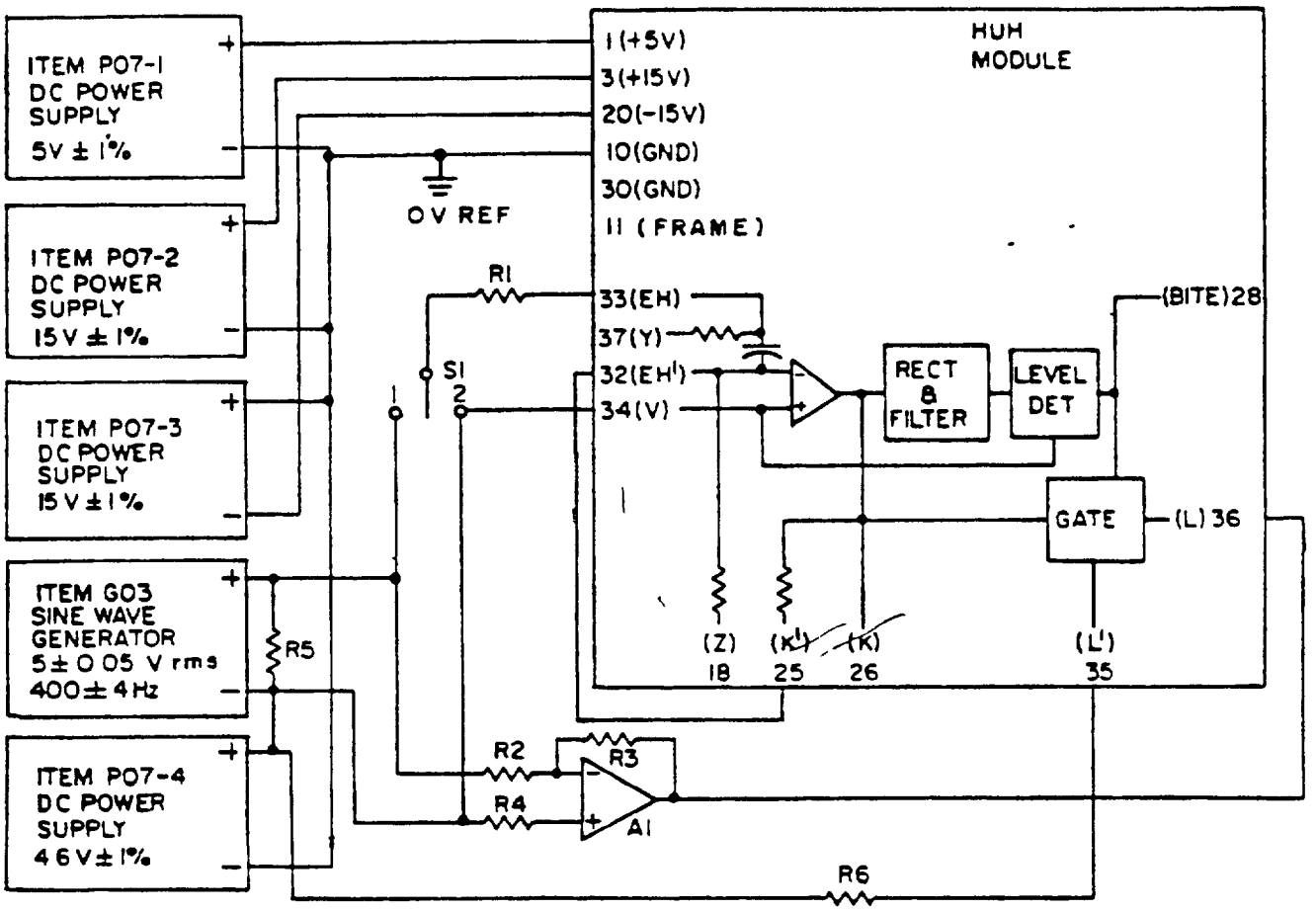
*TABLE V. Output short circuit current check.

| Test condition | Parameter | Switch position S1 | Switch ppsition S2 | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | | | Min | Max | | |
| D1 | Output short circuit current | 1 | 1 | --- | --- | --- | 8 |
| D2 | " | 2 | 1 | --- | --- | --- | " |
| D3 | " | 1 | 2 | --- | --- | --- | " |
| D4 | " | 2 | 2 | --- | --- | --- | " |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.

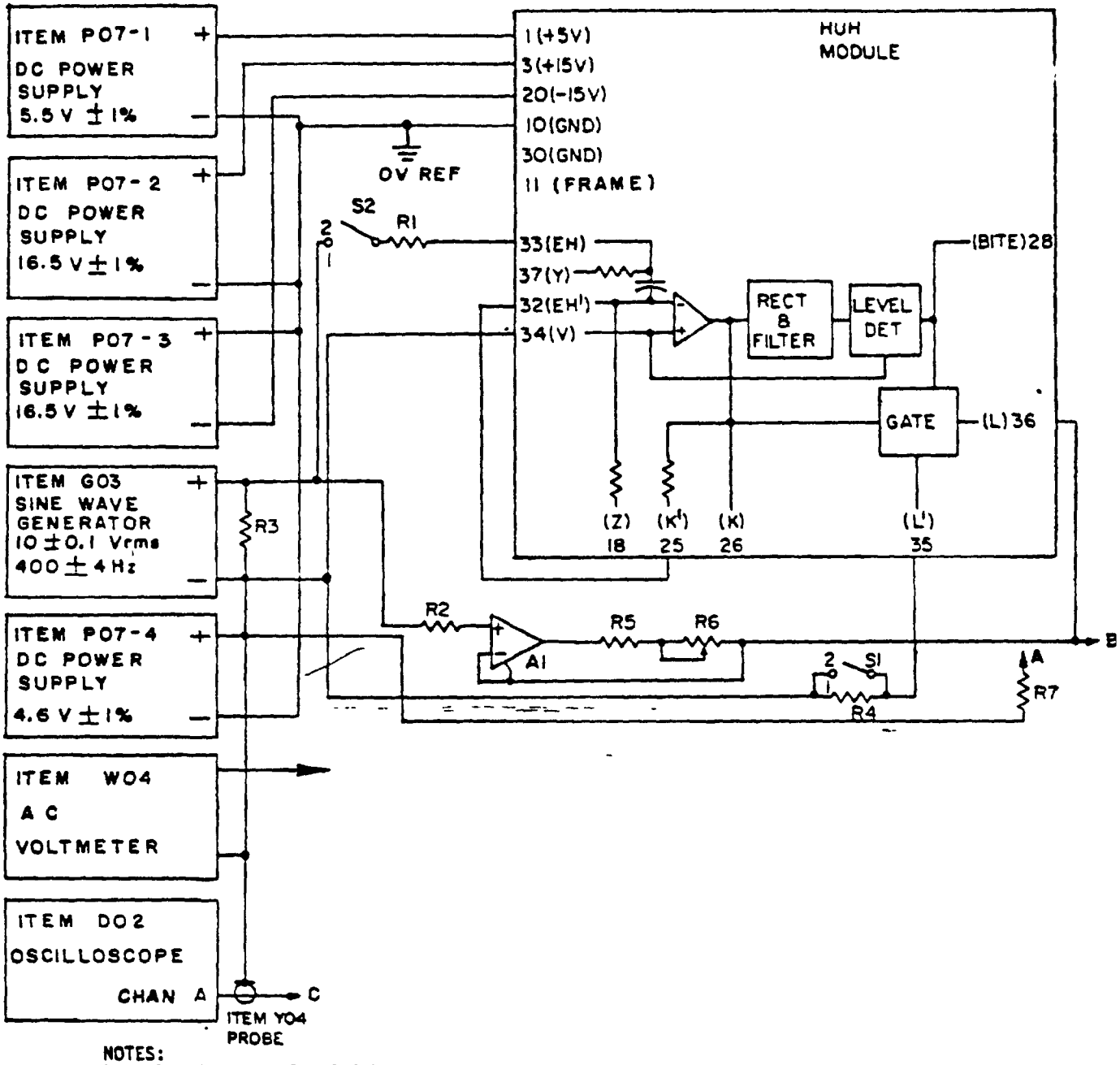



NOTES:
1. R1 = 45.3 kΩ ±1%, 1/8 W.
2. R2 = 1 kΩ ±1%, 1/8 W.
3. R3 = 10 kΩ ±1%, 1/8 W.
4. R4 = 5 kΩ ±1%, 1/8 W.
5. R5 = 600Ω ±10%, 1/8 W.
6. R6 = 20 kΩ ±1%, 1/8 W.
7. S1 = SPDT switch.
8. A1 = Fairchild type μA741 OpAmp, or equivalent.

FIGURE 7. Maximum input voltage test circuit for table IV.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.




NOTES:
1. R1 = 90.6 kΩ ±1%, 1/8 W.
2. R2 = 1 kΩ ±1%, 1/8 W.
3. R3 = 600Ω ±5%, 1 W.
4. R4 = 10Ω ±1%, 1/2 W.
5. R5 = 12Ω ±10%, 1/8 W.
6. R6 = 20Ω, 10 turn trim potentiometer.
7. R7 = 10Ω ±1%, 1/2 W.
8. S1 = S2 = SPST switch.
9. A1 = Fairchild μA791 OpAmp, or equivalent.
10. Connect point A to points B and C (module removed from socket).
11. Adjust R6 to obtain a reading of 500± 10 mVpk across R7.
12. Disconnect points A and C, insert module, and proceed with test.
13. Connect test for minimum of 10 seconds.
14. The current sense terminal of A1 shall be connected to its output as shown.
15. No readings are required for test conditions D1-D4.
16. The module passes if it passes the remaining tests.

* FIGURE 8. Output short circuit current check te circuit for table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.

TABLE VI. $V_S$ stress test.

| Test condition | Parameter | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|
| | | Min | Max | | |
| E | $V_S$ stress | -- | -- | -- | 9 |

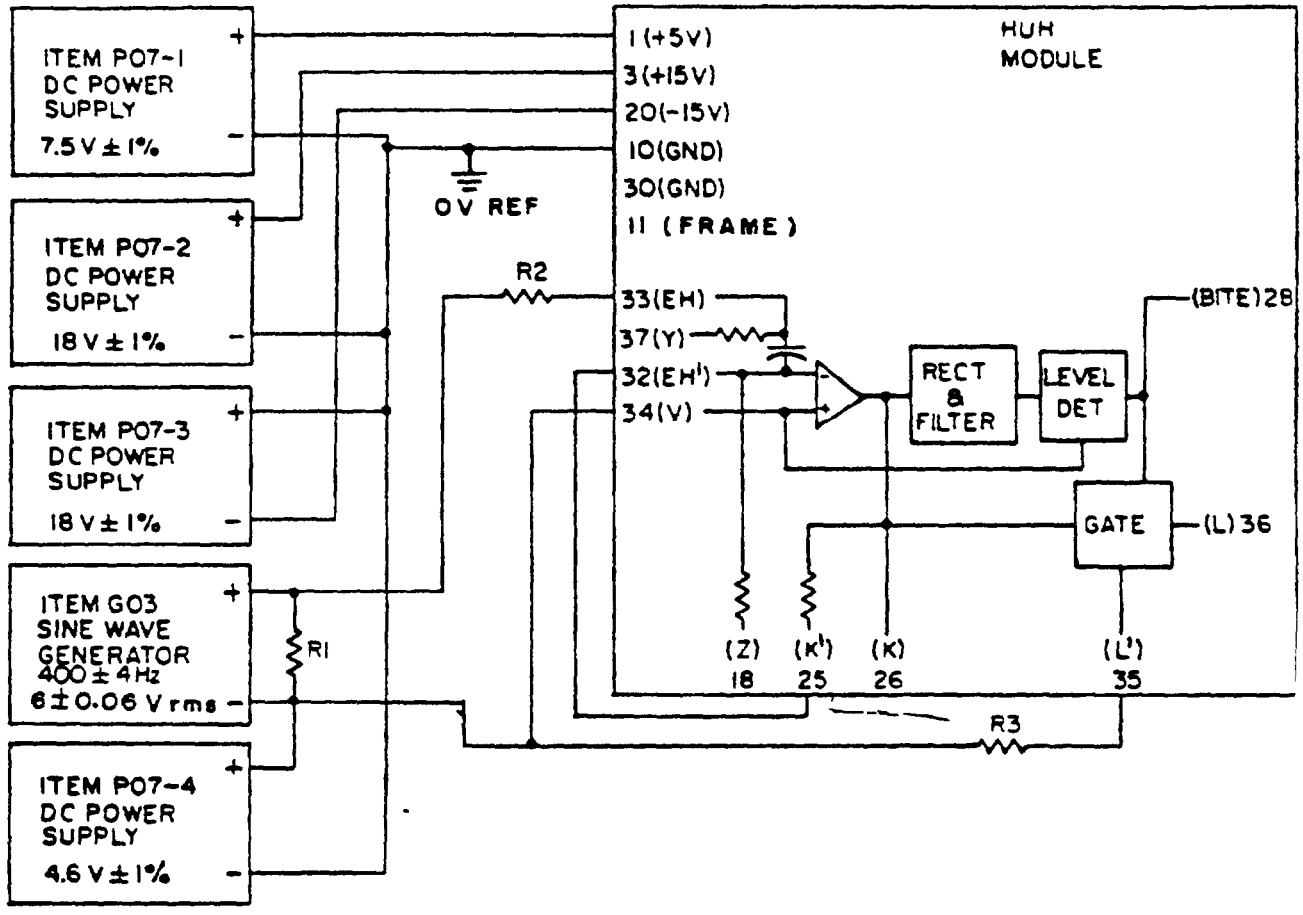
*TABLE VII. Crossover test.

| Test condition | Parameter | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| F1 | Crossover, course command | 36 | Open | A | 34 | 4.1 | 7.7 | V rms | 10 |
| F2 | Crossover, BITE check | " | " | 28 | 10 | 0.0 | 0.4 | V dc | " |
| F3 | Crossover, fine command | " | " | A | 34 | 2 | F1(-0.5) | V rms | " |
| F4 | Crossover, BITE check | " | " | 28 | 10 | 2.4 | 5.0 | V dc | " |
| F5 | Crossover, DC offset | Open | 36 | 35 | 36 | 0 | -3 | mV dc | " |
| F6 | Crossover, crosstalk | Open | 36 | 35 | 36 | 0 | 2 | mV rms | " |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.

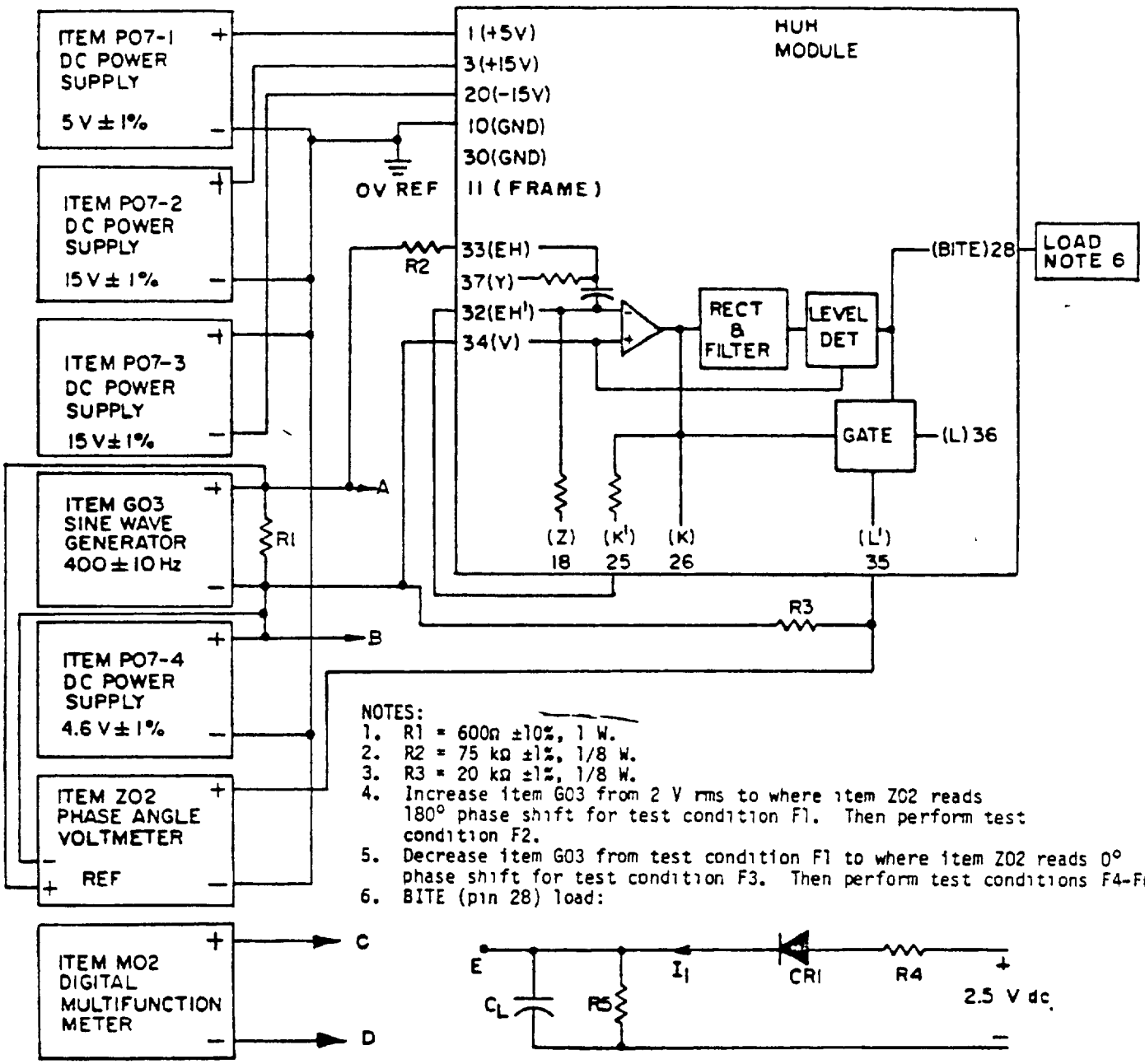



NOTES.
1. R1 = 600Ω ±10%, 1 W.
2. R2 = 45.3Ω ±1%, 1/8 W.
3. R3 = 20 kΩ ±1%, 1/8 W.
4. Leave module connected for a minimum of 15 seconds

FIGURE 9. $V_S$ stress test circuit for table VI.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.




NOTES:
1. R1 = 600Ω ±10%, 1 W.
2. R2 = 75 kΩ ±1%, 1/8 W.
3. R3 = 20 kΩ ±1%, 1/8 W.
4. Increase item GO3 from 2 V rms to where item ZO2 reads 180° phase shift for test condition F1. Then perform test condition F2.
5. Decrease item GO3 from test condition F1 to where item ZO2 reads 0° phase shift for test condition F3. Then perform test conditions F4-F6.
6. BITE (pin 28) load:




(a) R5 = 4.5 kΩ ±1%, 1/8 W.
(b) R4 = choose value such that $I_1$ = 6.5 mA 1% with point E at 0.4 V dc.
(c) CR1 = 1N914, or equivalent.
(d) $C_L$ = 100 pF 10%, mica, 20 WV dc.

FIGURE 10. Crossover test circuit for table VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.

*TABLE VIII. DC bias current (crossover switch) test.

| Test condition | Parameter | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|
| | | Min | Max | | |
| G1 | DC bias current, crossover switch | 0 | 750 | µA | 11 |
| G2 | DC bias current, crossover switch (BITE open circuit voltage) | 2.4 | 5.5 | V | 11 |

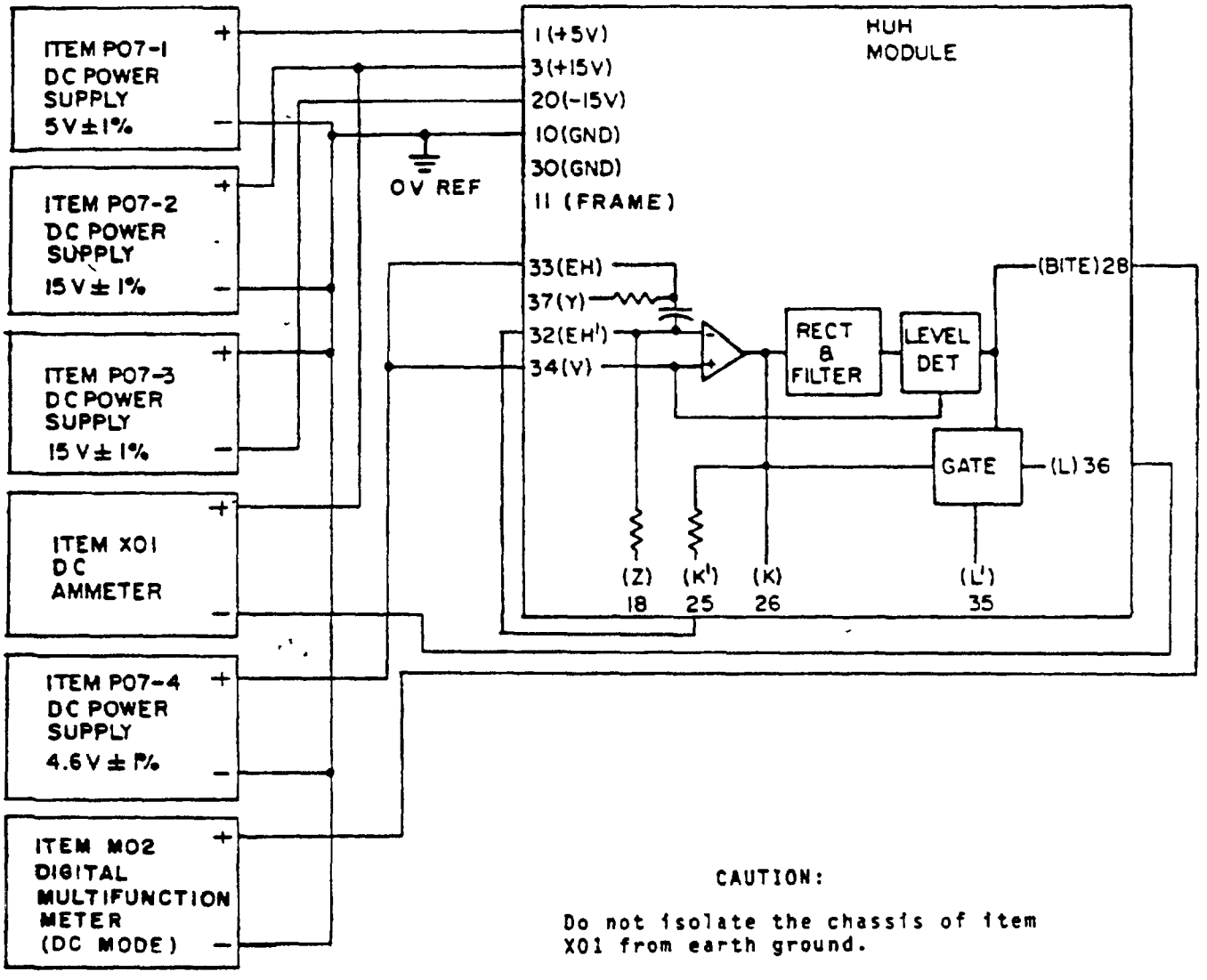
*TABLE IX. Resistance and Continuity test.

| Test condition | Parameter | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| H1 | Resistance | 33 | 37 | Open | Open | 108 | 132 | kΩ | 12 |
| H2 | Resistance | 32 | 18 | " | " | 27 | 33 | kΩ | " |
| H3 | Resistance | 25 | 26 | " | " | 67.5 | 82.5 | kΩ | " |
| H4 | Continuity | Open | Open | 11 | Frame | 0 | 1 | Ω | " |



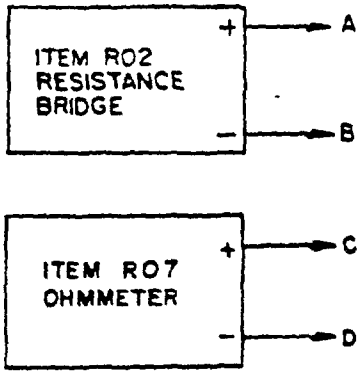


Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.




* FIGURE 11. DC bias current (crossover switch) test circuit for table VIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.




NOTE: For test condition H4, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of item RO7.

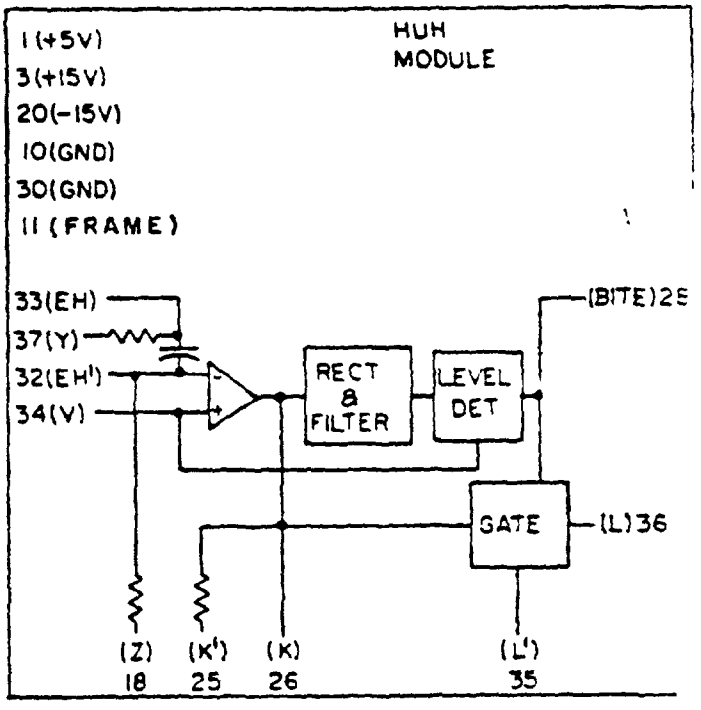



*FIGURE 12. Resistance and continuity test circuit for table IX.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.

TABLE X. Power supply filter capacitance test.

| Test condition | Parameter | Connect | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| J1 | Power supply filter capacitance | 33 | 32 | 12 | 18 | µF | 13 |
| J2 | " | 3 | 10 | 0.08 | 0.12 | " | " |
| J3 | " | 10 | 20 | 0.08 | 0.12 | " | " |

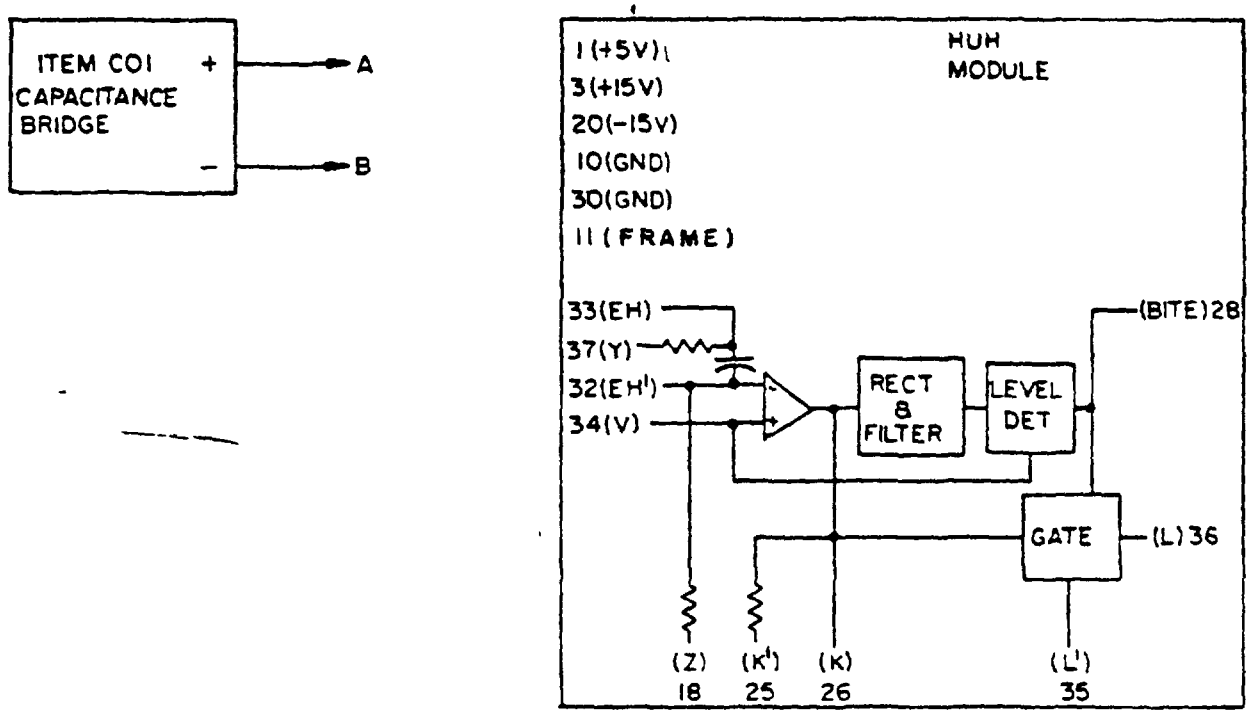
TABLE XI. Power supply current test.

| Test condition | Parameter | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|
| | | Min | Max | | |
| K1 | Power supply current (+ 5 V) | 0 | 15 | mA | 14 |
| K2 | Power supply current (+ 15 V) | " | " | " | " |
| K3 | Power supply current (- 15 V) | " | " | " | " |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.

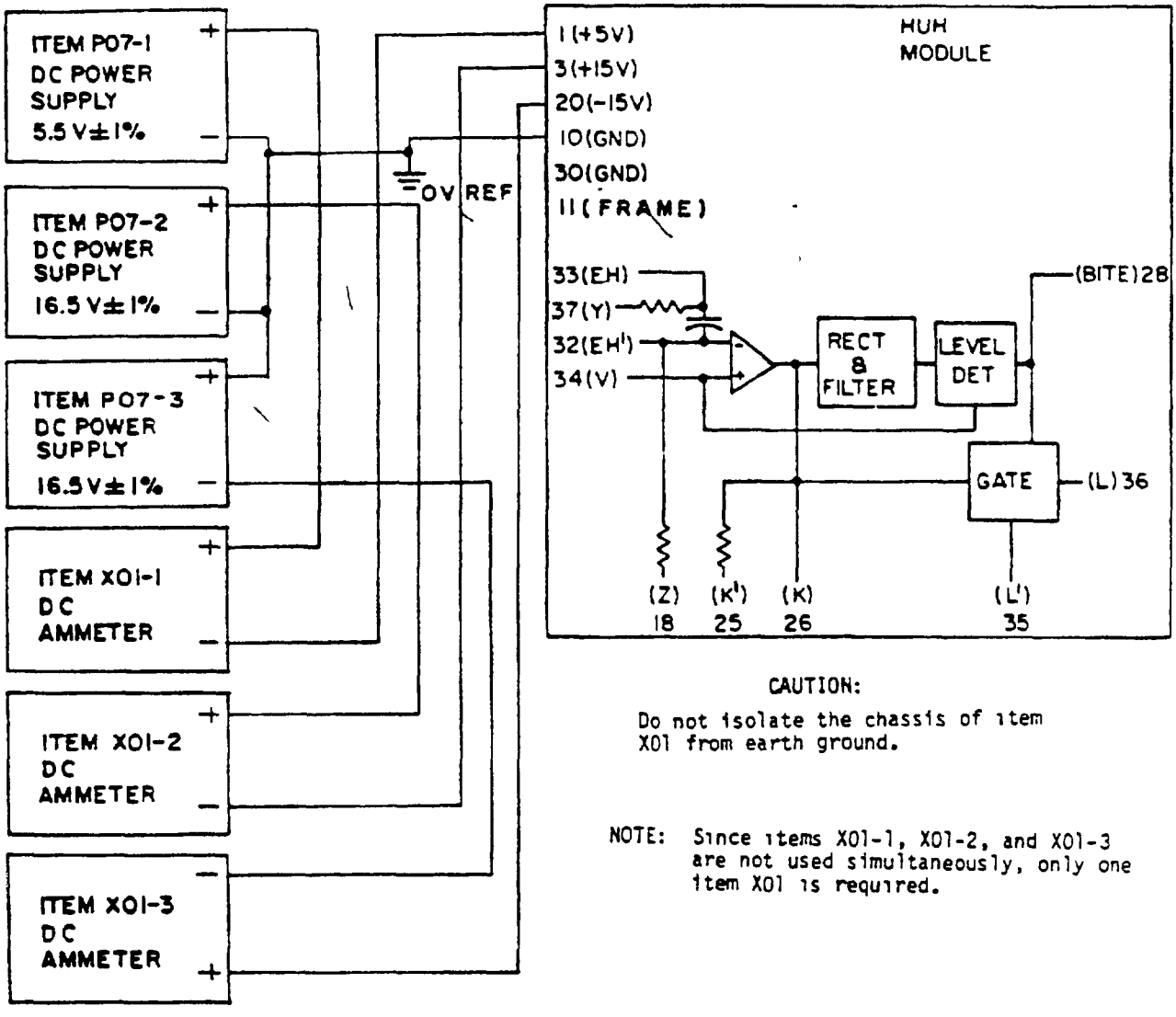



NOTE: Item C01 setup:
Frequency = 120 Hz for test condition J1.
Frequency = 1 kHz for test conditions J2 and J3.
$V_{pk}$ = 0.3 V ±1%.

*FIGURE 13. Power supply filter capacitance test circuit for table X.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.




CAUTION:

Do not isolate the chassis of item X01 from earth ground.

NOTE: Since items X01-1, X01-2, and X01-3 are not used simultaneously, only one item X01 is required.

*FIGURE 14. Power supply current test circuit for table XI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.

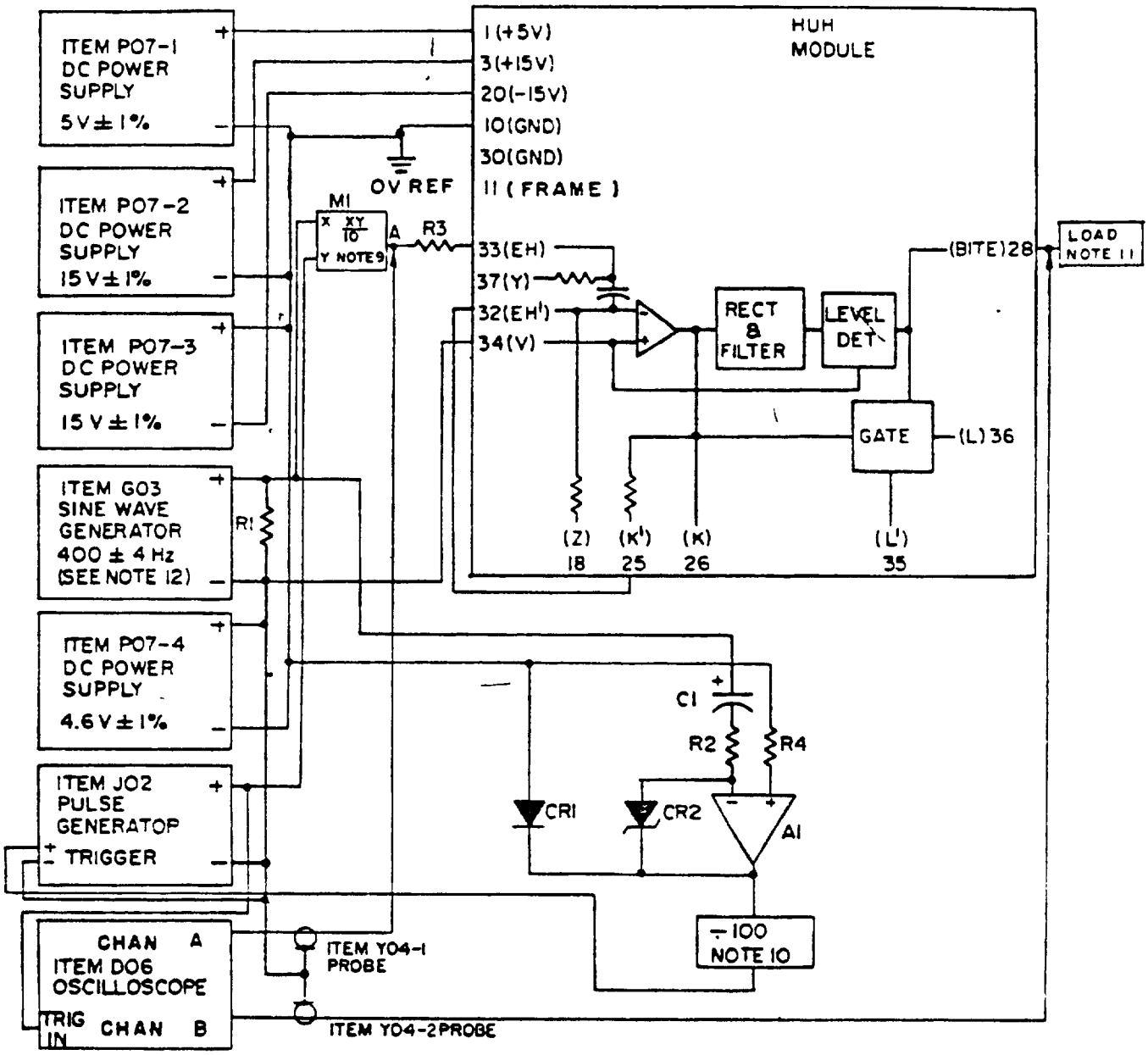
*TABLE XII. BITE propagation delay time ($t_{PHL}$ and $t_{PLH}$) and transition time ($t_{TLH}$ and $t_{THL}$) test.

| Test condition | Parameter | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|
| | | Min | Max | | |
| L1 | BITE propagation delay time ($t_{PHL}$) | 0 | 50 | ms | 15,16 |
| L2 | BITE propagation delay time ($t_{PLH}$) | 0 | 50 | ms | " |
| L3 | BITE transition time ($t_{TLH}$), $C_L$ = 100 pF | 0.1 | 200.0 | µs | " |
| L4 | BITE transition time ($t_{THL}$), $C_L$ = 100 pF | 0.1 | 200.0 | µs | " |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.

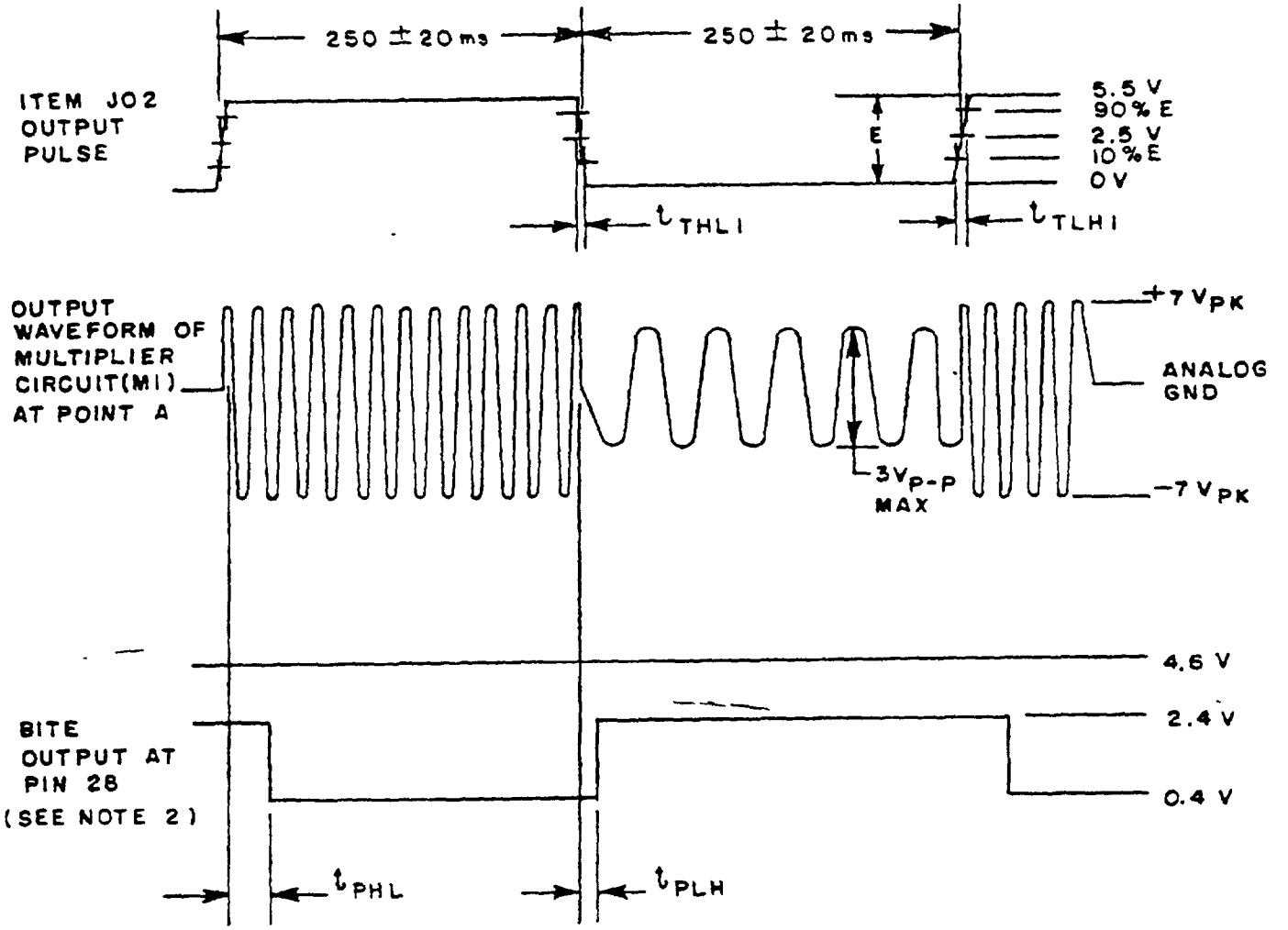



NOTES:

1. R1 = 600Ω ±5%, 1 W.
2. R2 = 5 kΩ ±10%, 1/8 W.
3. R3 = 45.3 kΩ ±1%, 1.8 W.
4. R4 = 1 kΩ ±1%, 1/8 W.
5. C1 = 100 µF ±10%, 50 WV dc, electrolytic.
6. CR1 = 1N914, or equivalent.
7. CR2 = 1N751A, or equivalent.
8. A1 = Fairchild µA741 OpAmp, or equivalent.
9. M1 = Analog Devices AD530LD multiplier, or equivalent.
10. Divide by 100 circuit consists of two 7490 decade counters, or equivalent, and one µA9614 line driver, or equivalent.
11. Load BITE (pin 28) per figure 10, note 6.
12. Using item D06, adjust the amplitude of item G03 to 7 $V_{pk}$ ±3% output at point A.

*FIGURE 15. BITE propagation delay time ($t_{PHL}$ and $t_{PLH}$) and transition time ($t_{TLH}$ and $t_{THL}$) test circuit for table XII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.




NOTES:
1. Item J02 output parameter: $t_{TLH1} = t_{THL1} \leq 0.1$ ms.
2. For test conditions L3 and L4, $t_{TLH}$ is measured between the 10% and 90% points and $t_{THL}$ is measured between the 90% and 10% points.

* FIGURE 16. BITE propagation delay time ($t_{PHL}$ and $t_{PLH}$) and transition time ($t_{TLH}$ and $t_{THL}$) test circuit waveforms for table XII.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-M-28787/225B(EC) | MODULES, STANDARD ELEC TRONIC SWITCH, S/D, MULTISPEED, ANALOG, KEY CODE HUH |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| | ☐ USER |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.

DD FORM 1426

TRUCTIONS In a continuing effort to make our standardization documents better, the DoD provides this form for use i mitting comments and suggestions for improvements. All users of military standardization documents are invited to pr estions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and d. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

*NOTE* This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

*(Fold along this line)*

*(Fold along this line)*

DEPARTMENT OF THE NAVY

Naval Electronic Systems Command
Washington, DC 20363

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20363
ATTN: ELEX 8111

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:47Z
Check the source to verify that this is the current version before use.